

MATH 213 – Homework 13

1. This assignment is about a study (Todorov, Mandisodka, Goren, & Hall, 2005, published in *Science*) on the appearance of candidates in political elections. The researchers who conducted the study wondered how well a group of people could correctly guess the outcomes of elections simply by choosing the candidates whose faces look more “competent.”

A group of study participants was shown pictures of the top two candidates from several elections. People in the group voted on who had the more competent-looking face. The researchers recorded which candidate won the majority vote of the participants as looking the most competent. One group of elections that the researchers used was the 32 U.S. Senate elections in 2004. For 23 of these elections, the candidate picked as having the more competent-looking face by the majority of study participants was actually the winner of the election.

A total skeptic might say that despite the data gathered in the study, having a group of people vote using the “competent face” method is really no better at guessing the winner of U.S. Senate elections than a coin flip.

- a. Write a null and alternate hypotheses we’ll use for our test, both in words and with the correct notation. Hint: are you looking at a mean or a proportion? Hint for the hint: is the variable in question categorical or numerical?

Null Hypothesis (H0): The competent-face method is no better than random guessing. That is, the proportion of correct predictions is 0.5.

Alternative Hypothesis (H1): The competent-face method is better than random guessing. That is, the proportion of correct predictions is greater than 0.5.

Notation version:

- **H0:** $p = 0.5$
- **H1:** $p > 0.5$

Where p = true proportion of elections correctly predicted by the competent-face method.

- b. In class, we used the “One-Proportion Inference” applet (<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to see how well the experimental data fit the skeptic’s theory (the “null model”). Use this applet to conduct a simulation, using at least 1000 repetitions, to assess how likely the skeptic’s “null model” (a coin flip for each prediction) is to produce data like what the researchers found in their study. Paste a screenshot of the results of your simulation here. **Make sure that you include a picture of the dotplot as well as a picture of your inputs for the simulation.**

One Proportion

Describe process:

Probability of heads:

Number of tosses:

Number of repetitions:

☒ Show animation

Total Repetitions = 1000

Choose statistic:

- ☒ Number of heads
☐ Proportion of heads

Count samples

As extreme as

Proportion of repetitions:
 $9 / 1000 = 0.0090$

Options:

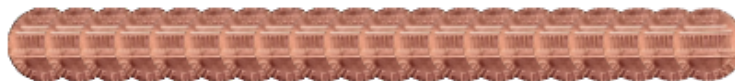
- ☐ Two-sided
☐ Exact Binomial
☐ Normal Approximation



Most recent results

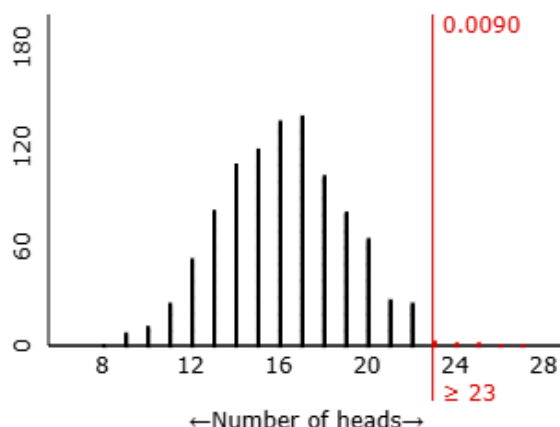


Number of Heads = 12



Number of Tails = 20

☐ Summary Statistics



☐ Show previous results

☐ Show sliders

- c. Describe the distribution of simulated correct predictions using the coin-flip method, which the applet displays as a dotplot. Address the center, spread, and shape of the distribution, writing in context about coin flips.

The distribution of simulated correct predictions using the coin flip method is symmetrical, centered around 16 with a spread ranging from 10 to 22. This makes sense because if predicting winners was like flipping a coin we could expect about 50% chance of correct predictions just by luck.

- d. Report also the “p-value,” which is the proportion of your repetitions where the coin-flip method predicted at least as many elections correctly as the “competent face” method did in the study. You can get this proportion in the applet by filling in an appropriate number for “as extreme as” and clicking the “count” button.

Pvalue = 0.009

- e. Judging from what you saw in the applet, is it plausible that the coin-flip method could produce data like what the researchers saw in their study? Explain your thinking.

Assuming there is no real advantage to the competent-face method and that it is no better than random guessing, getting 23 correct predictions out of 32 is extremely unusual. Out of 1000 simulated experiments using a coin-flip model, only 9 produced results as extreme as this one, resulting in a p-value of 0.009. Therefore, the probability of this result occurring due to chance or random variation is very small, which means that we have evidence to reject the null hypothesis. In other words, this study provides evidence that choosing the candidate with the more competent-looking face may be better than guessing when predicting election outcomes.